

Lab_V_4.2

- **Add 3 parameters to RAM module:-**
 - 1) **ADDR_WIDTH** of default value = 4
 - 2) **MEM_DEPTH** of default value = 8
 - 3) **MEM_WIDTH** of default value = 16
- **Replace the ports width with the new declared parameters**
- **Replace the 2D memory Width & Depth with the new declared parameters**

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```
module RAM (  
    input wire          write_enable,  
    input wire          clk,  
    input wire [2:0]    address,  
    input wire [15:0]   data_in,  
    output reg [15:0]   data_out  
);  
  
    // 2D Array  
    reg [15:0] memory [7:0];  
  
    always @(posedge clk) begin  
        if (write_enable) begin  
            memory[address] <= data_in;  
        end  
        else begin  
            data_out <= memory[address];  
        end  
    end  
  
endmodule
```